DS202
Inference human population history from individual whole genome sequences via PSMC

Anirudh Gupta Savyasachi Deval

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Problem Statement

- Can we recover **population history** from genome sequences alone?
- More specifically: can a **single diploid genome** tell us about past bottlenecks, expansions, and divergence?
- This is hard because recombination breaks the genome into many local genealogies.
- Today's motive:

PSMC → why it was powerful → what newer methods improve

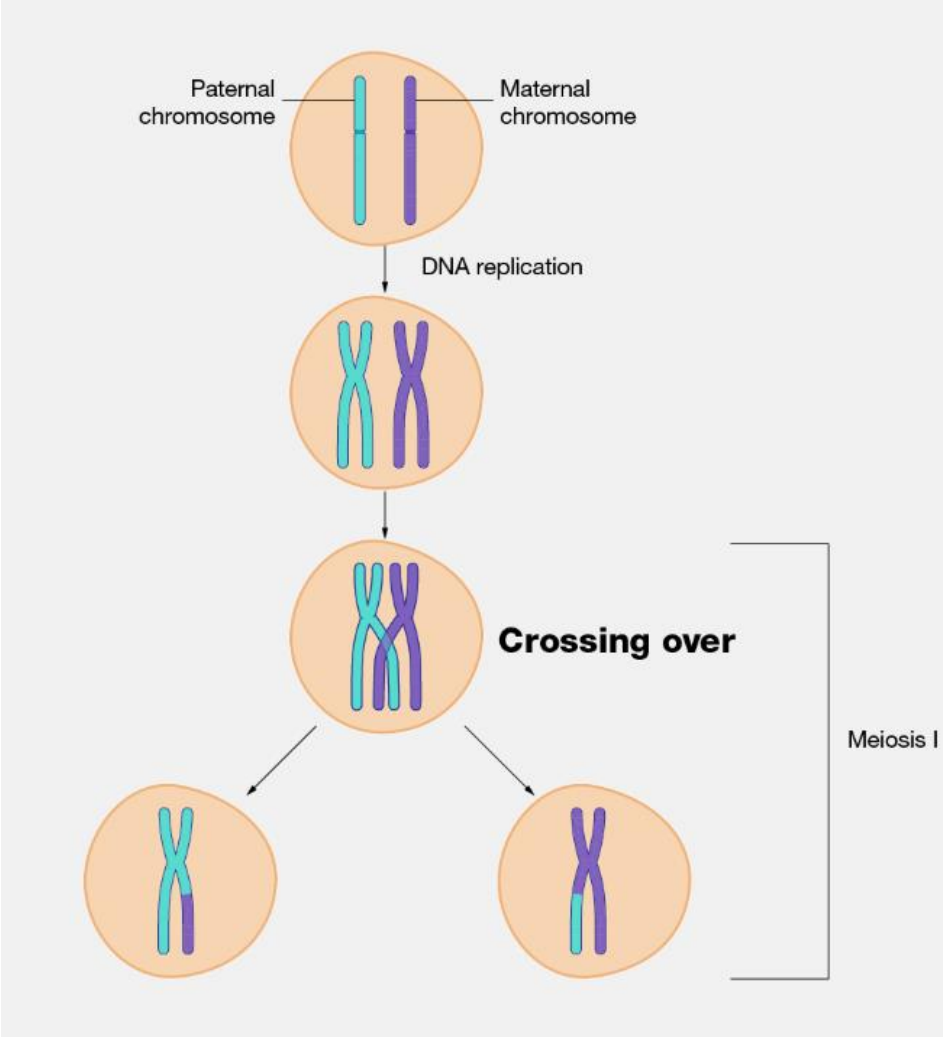
Coalescent Intuition

- Think backwards: two lineages eventually merge
- In a smaller population, coalescence will happen faster compared to a larger one.
- So the coalescence- time distribution, λ, can be given by

$$\lambda(t) = \frac{1}{N_e(t)}$$

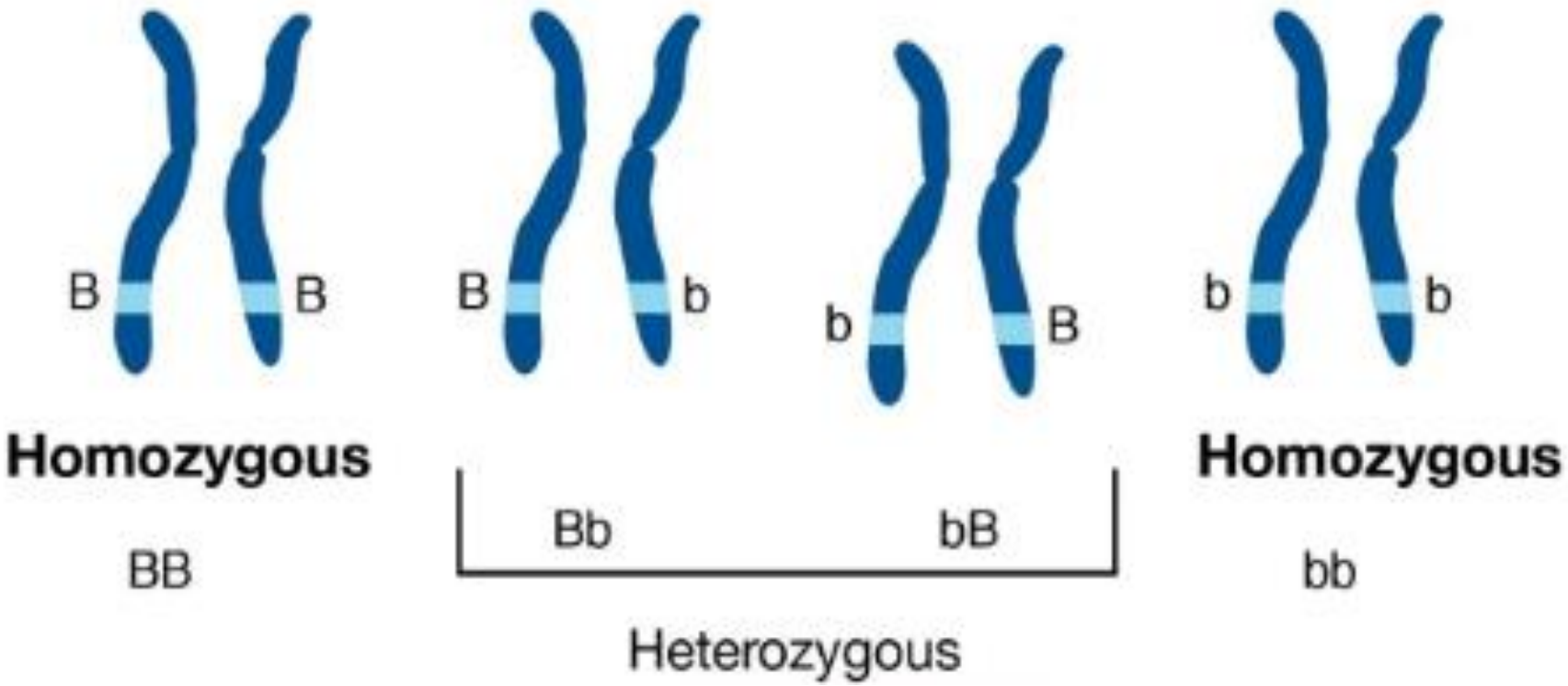
Why Recombination matters

- Without recombination, one genome gives only one genealogy
- With recombination different genome segments have different local trees
- This gives us many almost independent windows to the past
- PSMC exploits this by modelling changes in local ancestry along the genome.



Heterozygosity

- More heterogeneous regions usually mean older local coalescence
- Long Homozygous stretches suggest more recent common ancestry



PSMC : Pairwise Sequentially Markovian Coalescent

- It is a Hidden Markov Model (HMM) on the genome.
- Hidden State : discretized local TMRCA
- Observed Data: Whether each bin is Homozygous / Heterozygous / missing

Input representation

- A single, complete diploid whole-genome sequence. Basically a pair of string of same length, Example: ATCGGCGTTAA | ATCCGCGATAA
- These are divided into 100bp-bins
- We create another smaller representation for this in binary.
  - If all characters in each 100bp-bins are same => 0 (no mutation)
  - If there exists even a single mutation => 1 (mutation exists)
  - Example: for AAAA | AAAA => 0, for AAAA | AACA => 1
- This "binary" genome tells the HMM where mutations are clustered (suggesting an ancient ancestor) and where they are sparse (suggesting a recent ancestor).

Why size 100?

- The choice of **100-bp bins** is a strategic balance between statistical resolution and computational efficiency
- If bins were too large (e.g., 10,000 bp), multiple recombination events might occur within a single bin, blurring the signal.
- If bins were too small (e.g., 1 bp), almost every bin would be homozygous ('0'), making the HMM transition logic unstable.
- Computational complexity: 3-billion-base into 100-bp segments => ~30 million data points. This is small enough to run EM algorithm.

Hidden Markov Model states & parameters

- The hidden states in the HMM model are the time scales as output of the TMRCA value ranging from 10k years to 3 million years.
- The state is used in terms of time $\boldsymbol{t = g / 2*N_0}$, like if N_0 = 10000, g = 25, then t = 0.002
- The model typically uses **n=64** atomic time intervals in 1×4+25×2+1×4+1×6 pattern to reduce complexity, adjacent atomic intervals are often grouped together to share the same population-size parameter
- these are not equi-spaced but log scaled, so that they have more resolution on recent years than deep in past.

Hidden Markov Model states & parameters

- **θ**=$4N_0\mu$, where N_0 is the reference population size and μ is the mutation rate per site per generation. Determines the given probability of a bin being heterozygous ('1') given a specific TMRCA state.
- **Scaled Recombination Rate (ρ)** - Higher ρ means more frequent state transitions (recombination events).
- **Piecewise Constant Population Sizes (λ_i)**- $\lambda(t)=N_e(t)/N_0$ represents the relative effective population size during a specific time interval.
- **Generation Time(g)**: It represents the average time interval between the birth of an individual and the birth of their offspring, usually it is taken = 25 yrs for human population

Emission and Transition Probabilities

- **Emission Probability:** Defines the likelihood of observing a bin state (heterozygous '1' or homozygous '0') given a specific TMRCA state (t)

  $$e(x|t)=x(1-\exp(-\theta*t))+(1-x)\exp(-\theta*t) \text{ where } x \text{ in } \{0,1\}$$

- **Transition Probability:** Defines the likelihood of staying in the same TMRCA state or switching to another in the next bin.

  $$P(t|s)=(1-\exp(-\rho*t)).q(t|s) + \exp(-\rho*s)*x \text{ where } x = 1 \text{ iff } t=s$$

- The term $q(t|s)$ is the conditional probability of landing in state t, given that a recombination event has already occurred starting from state s. It is proportional to $\lambda(t)= N_e(t)/N_0$

Example Case Study

- Consider a small diploid Sequence

  AATTGGCCC AATGGCGT TTGGCCC

  AATTGGCCC AATGCGGT TTGGCCC

- Consider 100bp bins, so we can get the binary string as 0-1-0.
- Let θ=0.01(chance of mutation per bin), let's have only 2 bins for now which are t_1 = 0.1(recent) and t_2 = 1.0(ancient).
- We keep initial population N_0 = 10000 same for both bins.
- We keep ρ usually = $\theta/4$(as per PSMC standard), so ρ = 0.0025
- Since both bins have same population initially, $q(t_1|s) = q(t_2|s) = 0.5$ for both

Example Case Study(Continued)

- **For t_1(Recent): e(. | .) = P(. | .) => Emission Probability**
  - $P(1 \mid t_1) = 1 - \exp(-0.01 \times 0.1) \approx 0.001$ (1 in 1000 chance for mutation)
  - $P(0 \mid t_1) = \exp(-0.01 \times 0.1) \approx 0.999$
  - Recombination Probability: $1 - \exp(-\rho * t_1) = 1 - \exp(-0.0025 \times 0.1) \approx 0.00025$
- **For t_2(ancient)**
  - $P(1 \mid t_2) = 1 - \exp(-0.01 \times 1.0) \approx 0.010$ (1 in 100 chance for mutation)
  - $P(0 \mid t_2) = \exp(-0.01 \times 1.0) \approx 0.990$
  - Recombination Probability: $1 - \exp(-\rho * t_2) = 1 - \exp(-0.0025 \times 1.0) \approx 0.0025$

Example Case Study(Continued)

- For transition t_1->t_1
  - $P(t_1|t_1)=(1-0.00025)+(0.00025\times0.5)=0.99975+0.000125=0.999875$
- For Transition t_1->t_2
  - $P(t_2|t_1) = 0.00025\times q(t_2|\ t_1) = 0.00025\times0.5=0.000125$
- For Transition t_2->t_1
  - $P(t_2|t_1) = 0.0025\times q(t_1|t_2\ ) = 0.0025\times0.5=0.00125$
- For Transition t_2->t_2
  - $P(t_2|t_2) = (1-0.0025)+(0.0025\times0.5)=0.9975+0.00125=0.99875$

What Next? Motivation for EM

- To calculate the human population, our HMM needs the transition probabilities, which are dictated by the Effective Population Size ($N_e(t)$).
- We have assumed the $N_e(t)$ is same for all t initially. But we know the population will vary from the initial value.
- So we need to find a good way to update the value of $N_e(t)$ or simply $\lambda(t)$.
- **Main Idea:** It starts with a blind initial guess and iteratively updates the value of $\lambda(t)$ for each of the states, till it converges. Usually PSMC takes around 20 iterations for this.

Expectation (E-Step)

- The algorithm takes the *current guess* of the population history (N_e).
- It runs the HMM across the entire sequence of 0s and 1s.
- It calculates the **Expected** amount of time the genome spends in each of the 64 time intervals.

"Under this current guess, it looks like 15% of the genome comes from the ancient Time Bin 50."

Maximization (M-Step)

- It **Maximizes** the probability by adjusting the λ parameters. If a specific ancient time bin is visited very frequently by the HMM, the algorithm increases the population size for that era.

"If the genome really spends 15% of its time in Bin 50, what population size would make that the most logical outcome?"

EM (Continued) with Our Example

- We calculate the Raw Probability of Each Path
- For Path A: t_1 -> t_1 -> t_1

$$P(\text{Path A})=P(\text{start})\times e(0|t_1)\times P(t_1\rightarrow t_1)\times e(1|t_1)\times P(t_1\rightarrow t_1)\times e(0|t_1)$$

$$P(\text{Path A})=0.5\times 0.999\times 0.9998\times 0.001\times 0.9998\times 0.999$$

$$P(\text{Path A})\approx 0.000499$$

- For Path B: t_2 -> t_2 -> t_2

$$P(\text{Path B})=P(\text{start})\times e(0|t_2)\times P(t_2\rightarrow t_2)\times e(1|t_2)\times P(t_2\rightarrow t_2)\times e(0|t_2)$$

$$P(\text{Path B})=0.5\times 0.990\times 0.9987\times 0.010\times 0.9987\times 0.990$$

$$P(\text{Path B})\approx 0.004888$$

EM (Continued) with Our Example

- For Path C: t_1->t_2->t_1

$P(\text{Path C})=P(\text{start})\times e(0|t_1)\times P(t_1\rightarrow t_2)\times e(1|t_2)\times P(t_2\rightarrow t_1)\times e(0|t_1)$

$P(\text{Path C})=0.5\times 0.999\times 0.000125\times 0.01\times 0.00125\times 0.999$

$P(\text{Path C})\approx 7.79 \times 10^{-10}$ which is VERY SMALL

- The total probability of observing `0-1-0` is roughly the sum of these two paths A and B(Why?): 0.000499+0.004888=0.005387.
- Calculating for all paths is virtually impossible. Should we focus on paths with only single states???
  - Count will be 64^(30 million), (num states)^(length of binary string)

When PSMC changes state with high Probability

- Consider 0-0-0-1-1-1-1-1-0-0

If the emissions penalties is high for middle positions being in state $t_1$vs the transition probabilities, it makes sense to jump to t_2 in the middle.

- For 0-1-0, the model pays the transition penalty for jumping twice, however inside t_2, the emission probabilities of seeing 1's is much higher.

Forward Backward Algorithm

- **The Problem:** We cannot sum individual paths for a whole genome (30 million bins → too many paths!).
- The Forward-Backward algorithm calculates the exact "weight" (γ) for a state at a specific bin algebraically.
- **Forward Variable ($\alpha_i(t)$):** Probability of seeing observations *up to* bin i and landing in state t.
- **Backward Variable ($\beta_i(t)$):** Probability of seeing the *remaining* observations, given we are in state t at bin i.
- **The Weight Formula:** $\gamma_i(t)=\alpha_i(t)\times\beta_i(t)/P(O)$

Forward Backward Algorithm

Let's calculate $\alpha_2(t)$ for our heterozygous bin (1). (left to right)

- **Why?** It combines the probability of Bin 1 (0), the transition to Bin 2, and the emission of Bin 2 (1).
- **For t_1 (Recent):** $\alpha_2(t_1)=[\alpha_1(t_1)P(t_1 \rightarrow t_1)+\alpha_1(t_2)P(t_2 \rightarrow t_1)]\times e(1|t_1)$
  - $\alpha_2(t_1)\approx[0.4994+0.0006]\times 0.001=0.0005$
- **For t_2 (Ancient):** $\alpha_2(t_2)=[\alpha_1(t_1)P(t_1 \rightarrow t_2)+\alpha_1(t_2)P(t_2 \rightarrow t_2)]\times e(1|t_2)$
  - $\alpha_2(t_2)\approx[0.0001+0.4944]\times 0.010=0.0049$

- Note: Initial condition is 0.5, 0.5 as probabilities(and multiply with e)

Forward Backward Algorithm

Let's calculate $\beta_2(t)$ for our sequence.(right to left)

- **Why?** It is the probability of emitting the rest of the sequence (Bin 3 is 0), assuming we are currently in state t.
- **For t_1 (Recent):** $\beta_2(t_1)=P(t_1 \rightarrow t_1)e(0 \mid t_1)+P(t_1 \rightarrow t_2)e(0|t_2)$
- $\beta_2(t_1)\approx(0.999\times0.999)+(0.0001\times0.990)=0.999$
- **For t_2 (Ancient):** $\beta_2(t_2)=P(t_2 \rightarrow t_1)e(0 \mid t_1)+P(t_2 \rightarrow t_2)e(0|t_2)$
- $\beta_2(t_2)\approx(0.001\times0.999)+(0.998\times0.990)=0.990$

- Note: final condition, we take the Probabilities as 1.0 (no emission multiplied)

EM(Continued): Obtain probabilities

Now we fuse the past (α) and the future (β) to get the total path probability.

- **Total P(O):** $(\alpha_2(t_1)\times\beta_2(t_1))+(\alpha_2(t_2)\times\beta_2(t_2))=0.0005+0.0048=0.0053$
- **Weight of t_1 (Recent):** $\gamma_2(t_1)=0.0005/0.0053 \approx 9.3\%$
- **Weight of t_2 (Ancient):** $\gamma_2(t_2)=0.0048/0.0053 \approx 90.7\%$

EM (Continued) with Our Example

Bin 1:

- $\gamma_1(t_1)=(0.4995\times0.001001)/0.005395=0.0927$ **(9.27%)**
- $\gamma_1(t_2)=(0.4950\times0.009888)/0.005395=0.9073$ **(90.73%)**

Bin 2:

- $\gamma_2(t_1)=(0.000500\times0.9990)/0.005395=0.0926$ **(9.26%)**
- $\gamma_2(t_2)=(0.004945\times0.9900)/0.005395=0.9074$ **(90.74%)**

Bin 3:

- $\gamma_3(t_1)=(0.000506\times1)/0.005395=0.0938$ **(9.38%)**
- $\gamma_3(t_2)=(0.004889\times1)/0.005395=0.9062$ **(90.62%)**

EM (Continued) with Our Example

- Final Expectation (E): We sum up the fractional weights (γ) across all bins.

$$E[t|\text{Bin } i]=(1 \times \gamma_i(t))+(0 \times (1-\gamma_i(t)))$$

$$E[t] =\sum \gamma_i(t) \text{ from } i = 1 \text{ to } L$$

$$=E(t_1)=0.0927+0.0926+0.0938=0.279$$

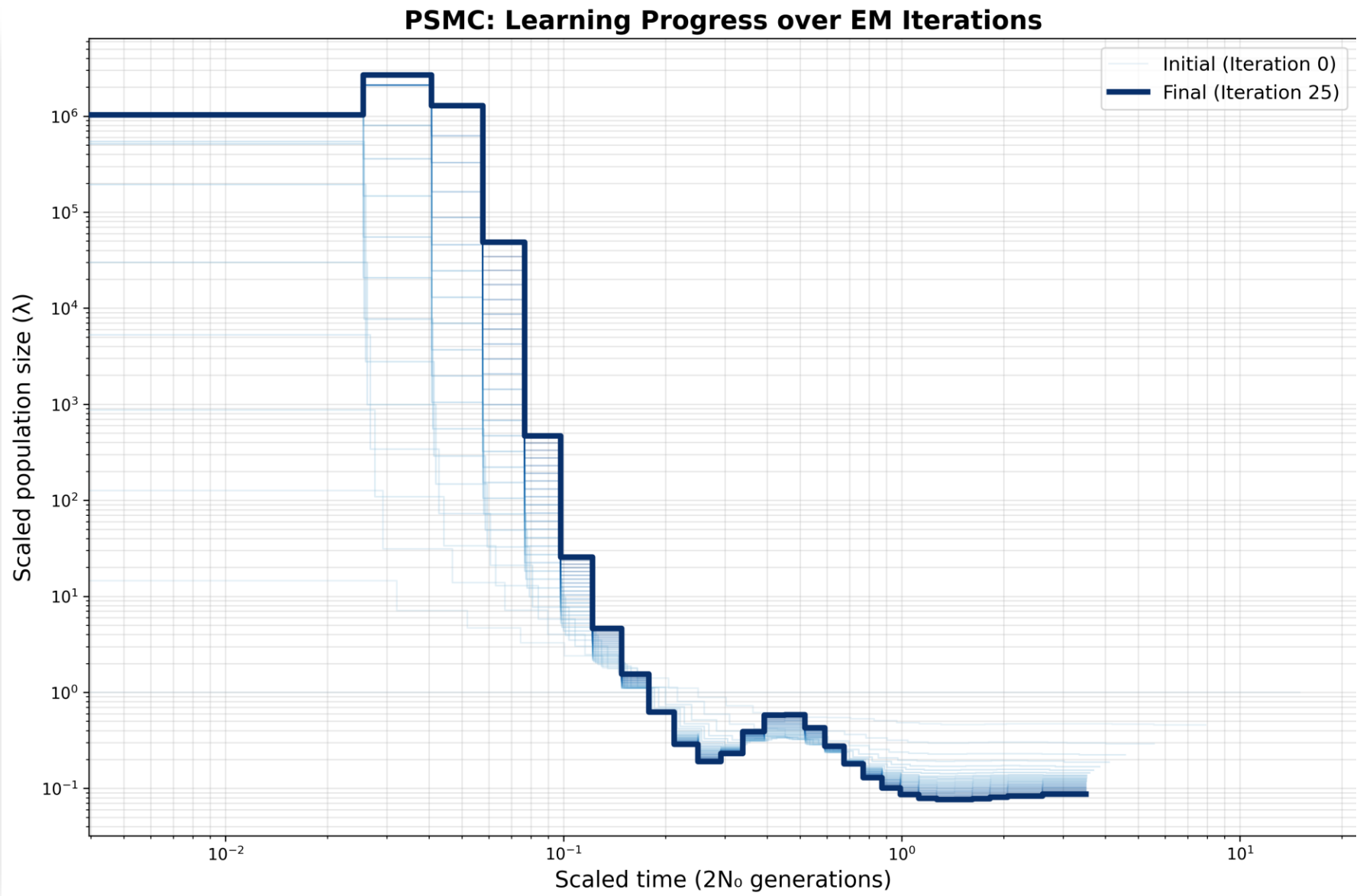
$$=E(t_2)=0.9073+0.9074+0.9062=2.721$$

M Step: Update λ and get New N_e

Update relative population size λ(t) based on the E-step's total expectations: $E(t_1)=0.279$, $E(t_2)=2.721$.

- The new λ values are updated proportionally to the total expected bins (3 total).
- **New Recent Pop (λ_1):** 0.279/3 ≈9.3%. λ_1 drops from 1.0 to ≈0.186.
- **New Ancient Pop (λ_2):** 32.721 ≈90.7%. λ_2 jumps from 1.0 to ≈1.814.

Iteration 1 has successfully detected a massive historical bottleneck.


PSMC: Learning Progress over EM Iterations
Initial (Iteration 0)
Final (Iteration 25)
Scaled population size (λ)
10^{6}
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-1}
10^{0}
10^{1}
Scaled time ($2N_0$ generations)

Final Population

- Let after some iterations, we get λ_1 =0.05 and λ_2=1.95,
- final $N_e(t_1) = \lambda_1 * N_0$= 10000*0.05=500
- Final $N_e(t_2) = \lambda_2 * N_0$= 10000*1.95=19500

Since TMRCA is independent of the population changes, we have that TMRCA for a state t_k is given by $t_k * 2N_0 * g$

For t_1 = 0.1, TMRCA = 0.1*20000*25=50000 = 50K yrs

For t_2 = 1.0, TMRCA = 1.0*20000*25=5,00,000 = 0.5 million yrs

Interpretting PSMC graphs

- The X axis denotes the time bins in years which are non-uniform as discussed (usually scaled in $2*N_0$), Y axis is population size.
- Two paths differ where major recombinations events occur like migration, cross-breeding, etc.
- Due to this, 2 paths of any population diverge and population of both differ significantly.

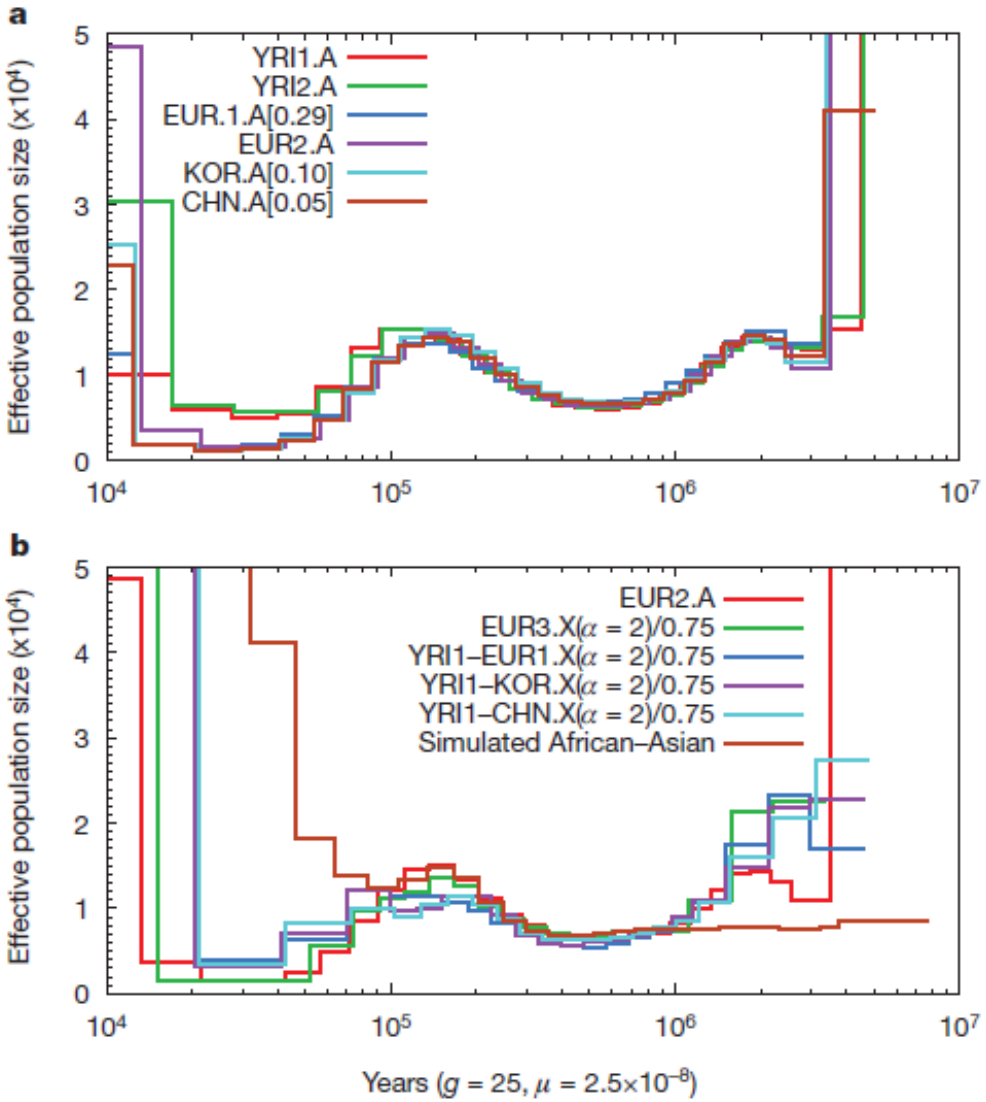
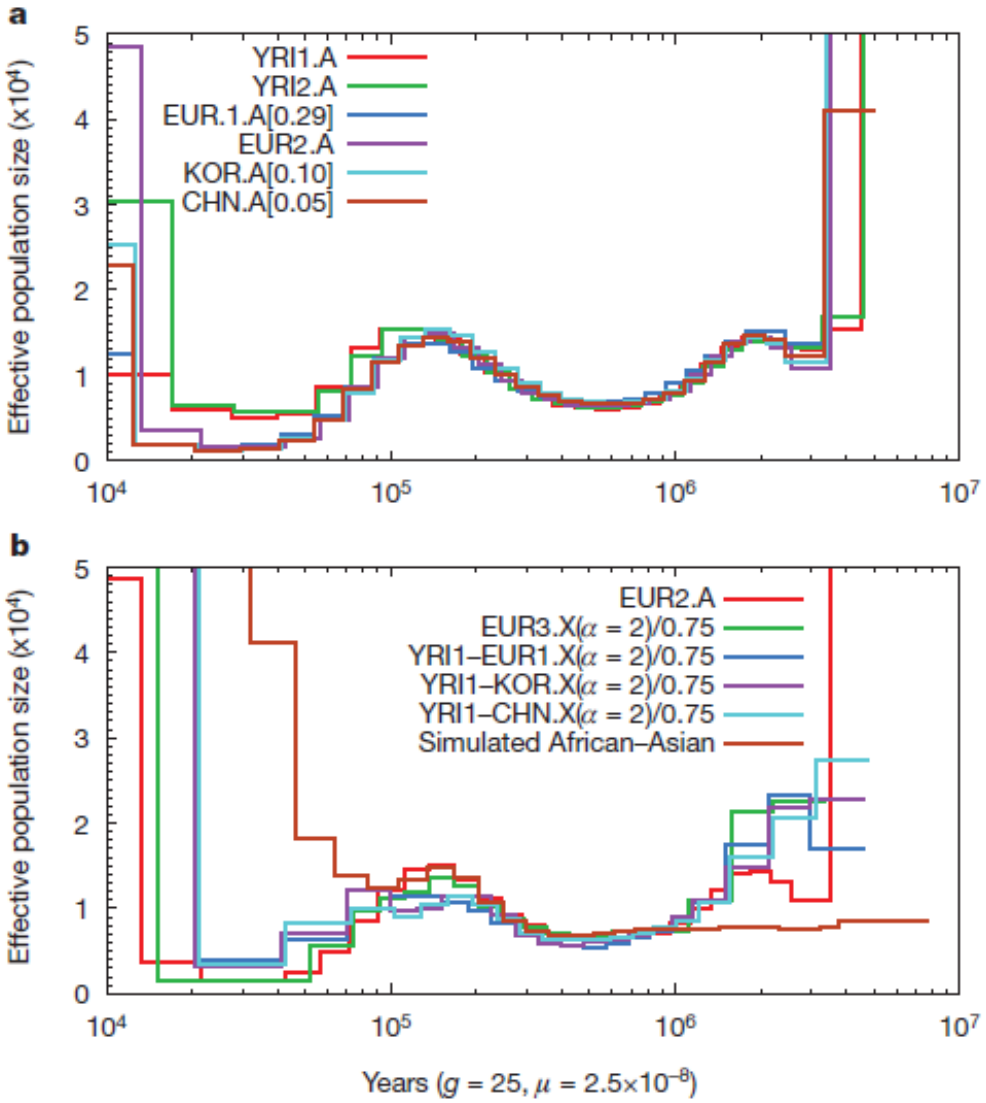


Figure 3 | PSMC estimate on real data. **a**, Population sizes inferred from autosomes of six individuals. 5%, 10% and 29% of heterozygotes are assumed to be missing in CHN.A, KOR.A and EUR1.A, respectively. **b**, Population sizes inferred from male-combined X chromosomes and the simulated African–Asian combined sequences from the best-fit model in ref. 21. Sizes inferred from X-chromosome data are scaled by 4/3. The neutral mutation rate on X, which is used in time-scaling, is estimated with the ratio of male-to-female mutation rate, α, equal to 2 (see Methods).

Interpretting PSMC graphs

- We can see in Figure B, the plots are in between 10^5 and 10^6 are similar, hence similar ancestry is observed with no major recombinations are seen and can be inferred all originating from African population gene near 10^6-10^7 yrs
- At 10^5, all populations paths differ suggesting recombinations events like migration. We see China and Korea are similar and all European takes similar path, which means groups migrated different places and hence took a similar path.



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Simulated data Experiments from the paper Setup

- Li-Durbin simulated 100 sequences of 30 Mb with an out of Africa bottleneck plus expansion
- Tested variable mutation rates and recombination hotspots.
- Simulated data were binned exactly like real data.

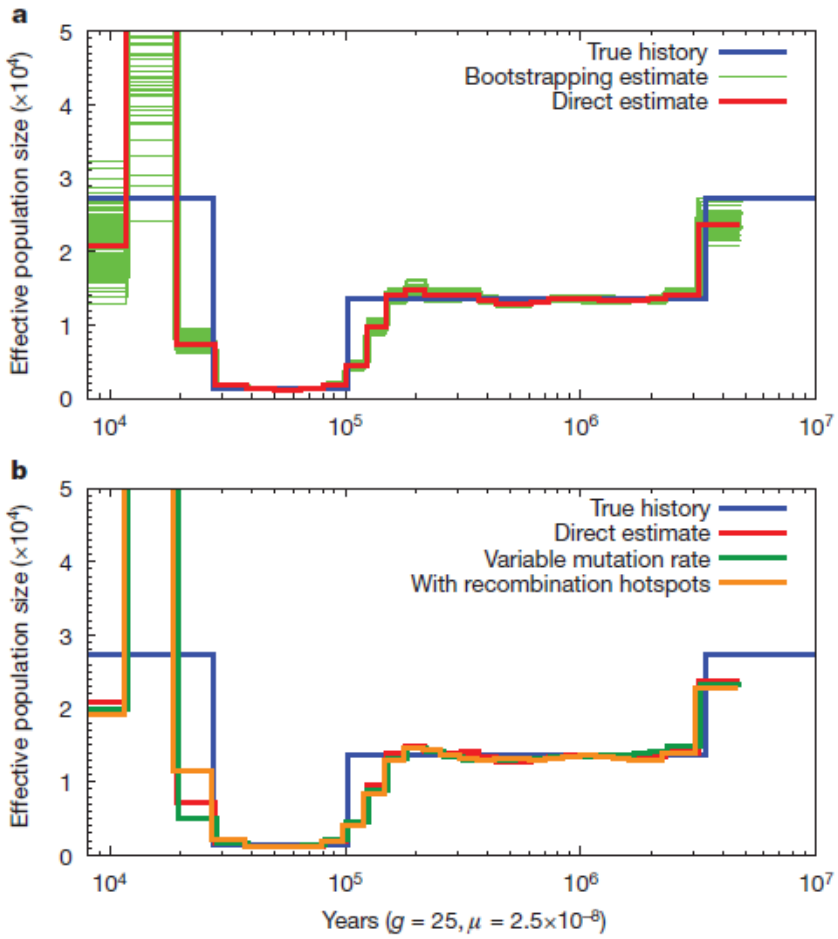


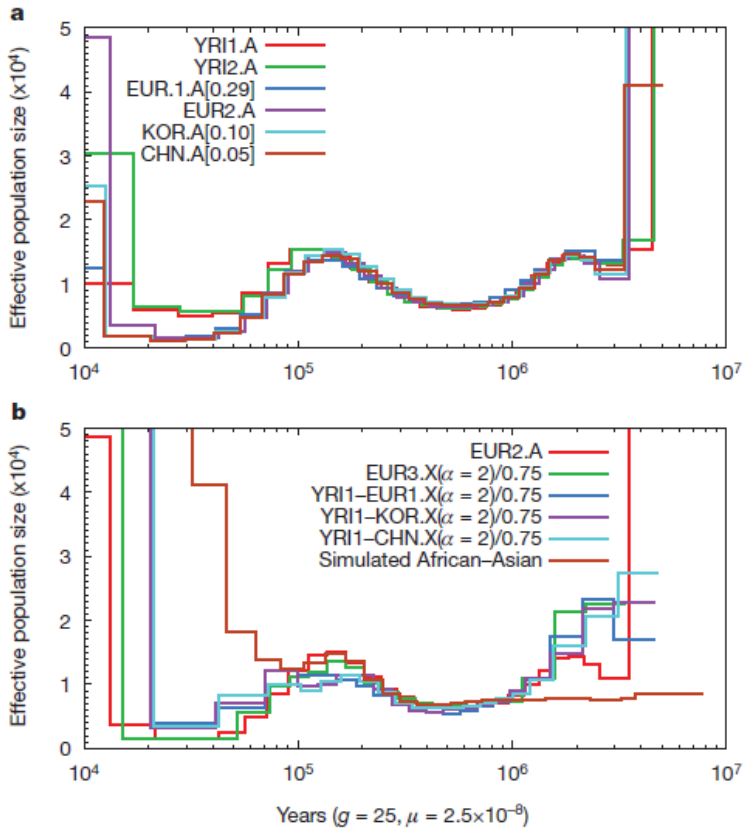
Figure 2 | PSMC estimate on simulated data. **a**, PSMC estimate on data simulated by msHOT. The blue curve is the population-size history used in simulation; the red curve is the PSMC estimate on the originally simulated sequence; the 100 thin green curves are the PSMC estimates on 100 sequences randomly resampled from the original sequence. **b**, PSMC estimate on data with a variable mutation rate or with hotspots. g, generation time; μ, mutation rate.

Simulated data Experiments from the paper Results

- Best performances was roughly between 20 kya and 3 Mya
- Remained robust under mutation rate variation and recombination hotspots
- Struggles At very recent and ancient because of too few informative events remain there.
- Sudden changes in real data are often represented as smooth transition.

Results and findings from the paper (1): Non-African Bottleneck

- European and Chinese histories are very similar before about 10-20 Kya
- Both show a severe bottleneck around 10-60 kya indicating the small population that migrated out of Africa
- African Populations show a milder bottleneck and recover earlier providing an evidence that African populations retained diversity
- This supports the Out of Africa Theory



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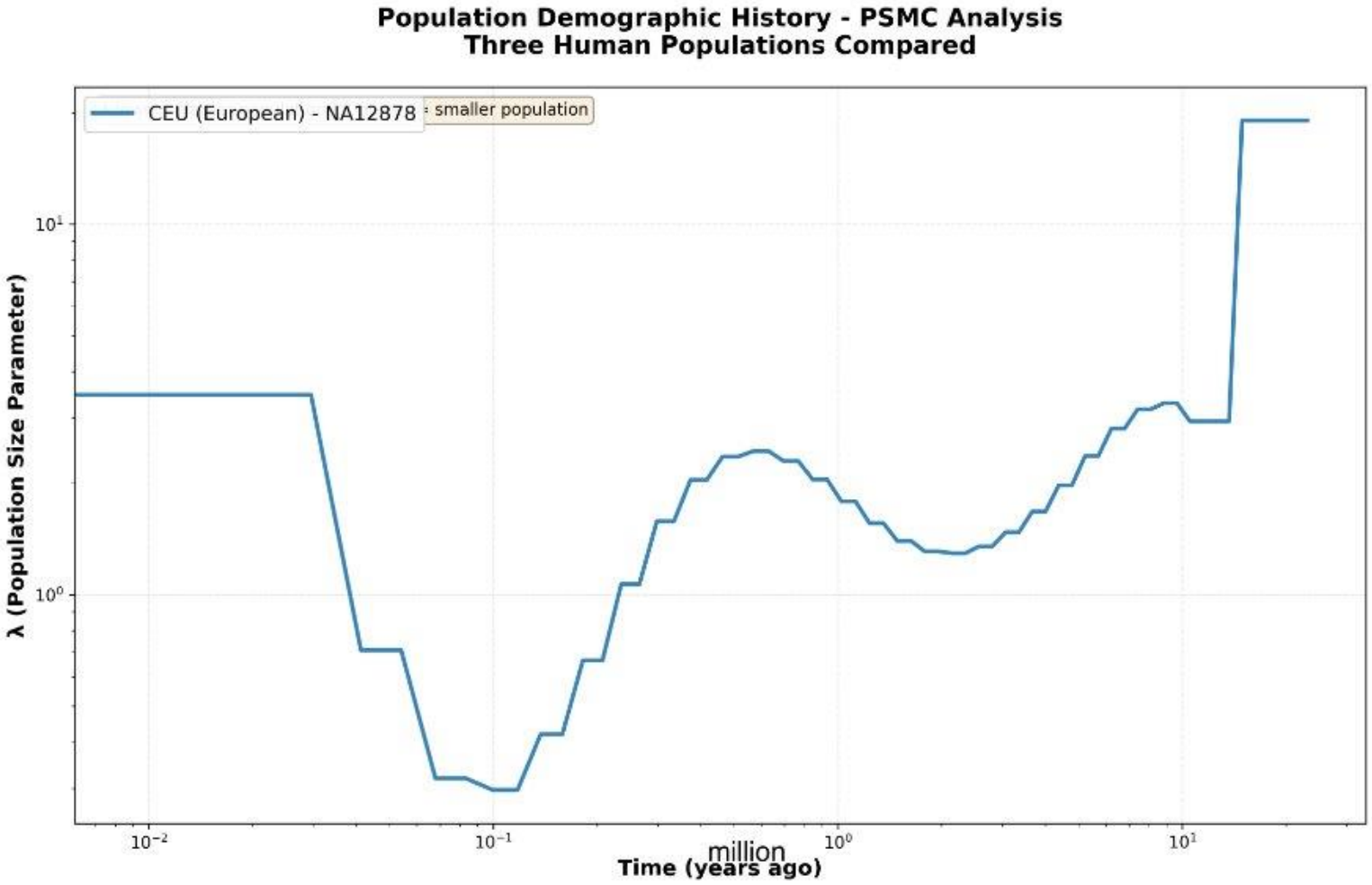
Our experiment for NA12878 (European)



Results and findings from the paper (2): Deeper time interpretation

- All three populations show an elevated inferred N_e between roughly 60 and 250 kya
- This may reflect population substructure
- Authors also infer that differentiation of modern humans begin around 100-200 kya
- It is noteworthy that we are performing the analysis on the genome and hence the absence of the same species in ancient time frame doesn't prevent us from inferring the ancestral information.

Caveats in interpretations

- The inferred curve is an **effective** size history, not a literal census count.
- Population structure can mimic size changes.
- Absolute dates depend on mutation rate and generation time assumptions.
- PSMC is limited for very recent history and for sudden demographic events.

PSMC conclusion

- PSMC uses only **one diploid sample**.
- It works with a simplified piecewise-constant hidden-state model.
- Extending to multiple genomes means a much more complex hidden state space: **trees / ARGs**.
- This motivates methods like **SINGER** and **PHLASH**.

PHLASH: A Bayesian extension

- PHLASH returns a **Bayesian posterior over population size histories**.
- able to use both **linkage** and **frequency-spectrum** information
- Standard PSMC-like methods discretize time using a **fixed grid**.
- That choice can bias the shape of the inferred size-history curve.
- PHLASH instead places a prior over size histories and **randomly discretizes time**.
- Averaging many posterior draws produces a smoother, more adaptive estimate.

PHLASH MATH

- Overall composite log-likelihood:

$$\log \mathbb{P}_{\text{PHLASH}}(g \mid \eta) = \sum_{i=1}^{N} \log \mathbb{P}_{\text{PSMC}}(g_i \mid \eta) - d_{KL}(\xi_\eta \,\|\, \xi_g) - d_\Sigma(LD_\eta, LD_g)$$

- First term: linkage-based PSMC information
- Second term: SFS fit (Side frequency spectrum);
  - How many mutations are rare vs common
- Third term: LD fit (Linkage Disequilibrium)
  - Measures how often alleles are inherited together

OVERVIEW

- For genomes g1,…,gn , PHLASH uses a **composite likelihood**:

$$\mathbb{P}(g \mid \eta) \approx \prod_{i=1}^{n} \mathbb{P}(g_i \mid \eta)$$

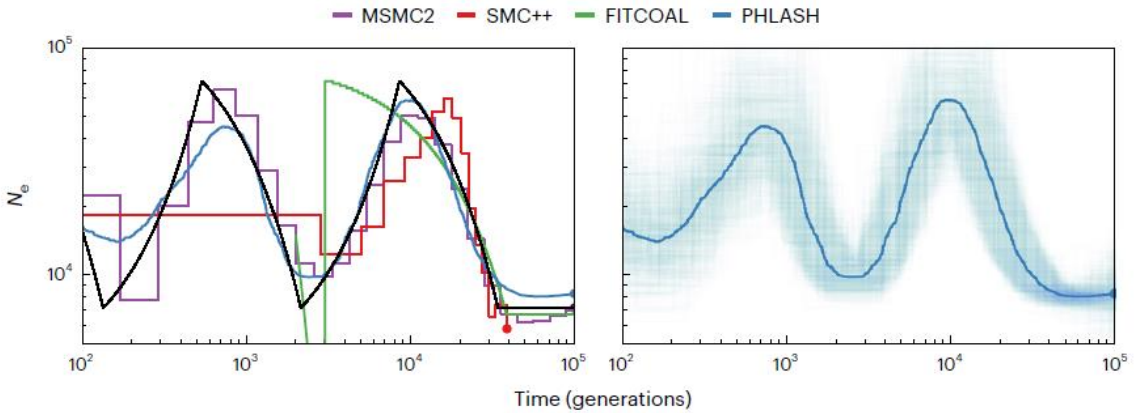
- Each term is a PSMC-like single-genome likelihood.
- This is computationally simpler than a full joint model.
- It lets PHLASH scale to much larger sample sizes.

PHLASH Computation and optimisation

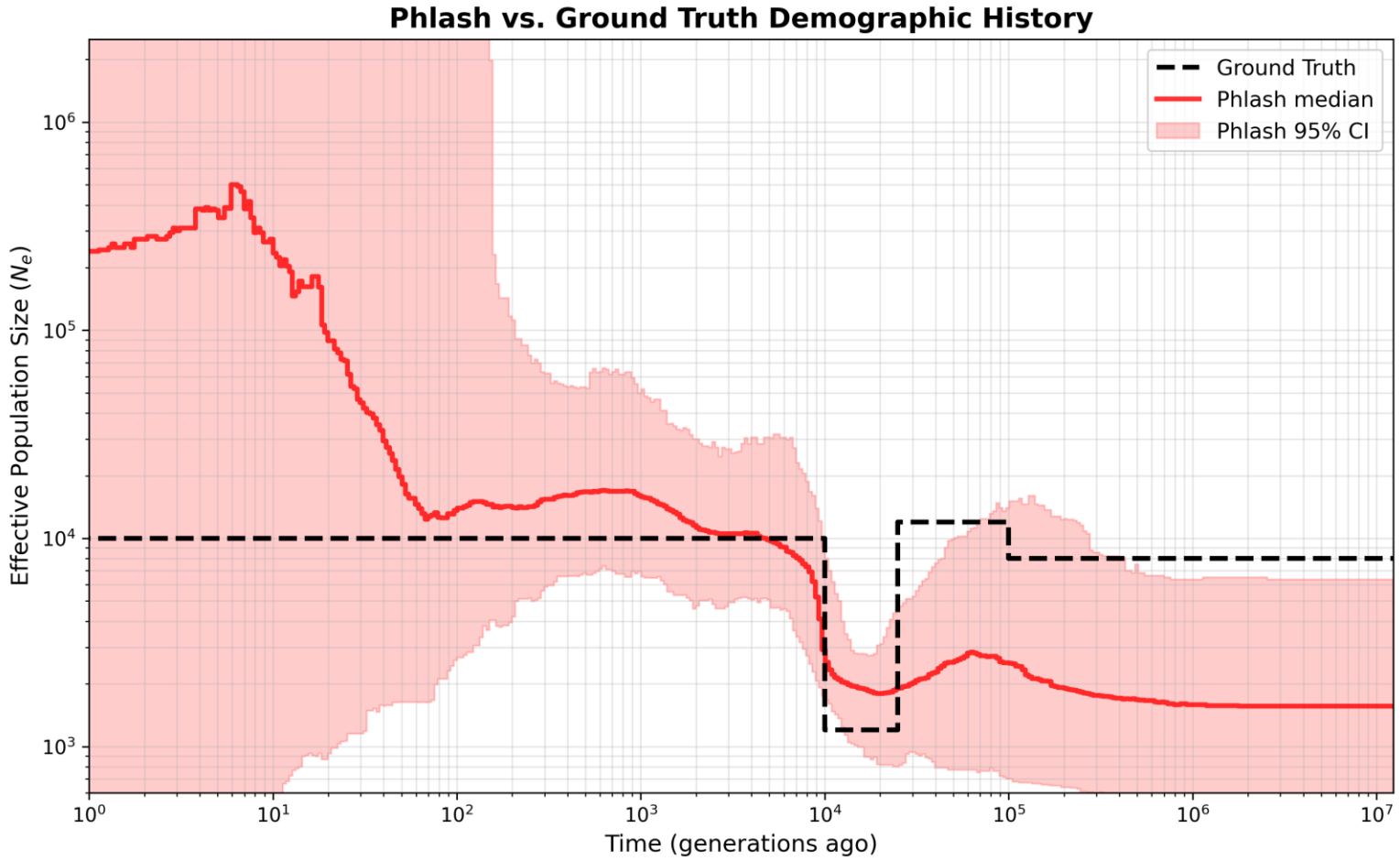
- Bayesian inference usually needs gradients of the log-likelihood.
- PHLASH introduces a new method for computing the **score function** efficiently.
- The paper states this gradient can be computed at the **same cost as the log-likelihood**.
- The software also supports **GPU acceleration**.

PHLASH results

- On simulated data, PHLASH often had lower error than SMC++, MSMC2, and FITCOAL.
- It achieved the highest accuracy in **22 of 36 scenarios** in the benchmark.
- It returns **credible intervals / posterior uncertainty**, not just a point estimate.
- At large sample sizes, it remained computationally practical when others struggled.



PHLASH on Simulated Dataset



PHLASH Scalablity

- For n=1, most methods had comparable CPU time.
- For larger sample sizes, PHLASH required substantially less CPU time and memory than MSMC2/SMC++ among HMM-style methods.
- For n=1000, the paper reports that **only PHLASH** could run within the allocated resources.
- This makes it attractive for modern large-scale datasets.

SINGER: ARG based approach

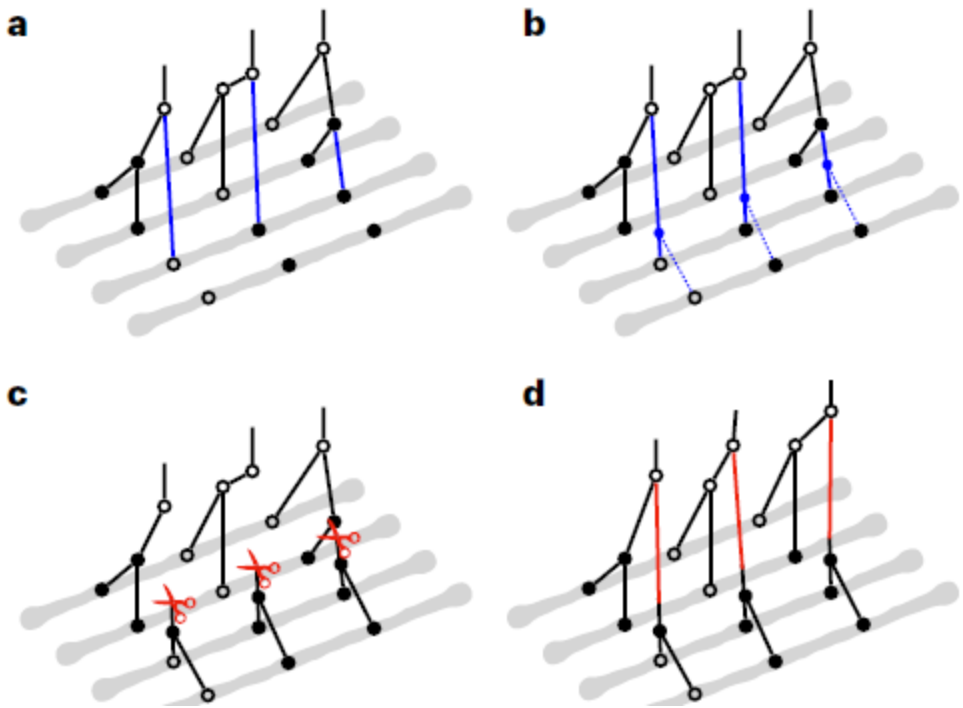
- SINGER targets inference of the **Ancestral Recombination Graph (ARG)** for many genomes.
- Modern scalable ARG methods often lose accuracy, ignore uncertainty, or assume oversimplified priors.
- SINGER is designed to be **Bayesian, faster, more accurate, and uncertainty-aware**.
- It works on **phased whole-genome sequence data**.

What is an ARG

- With recombination, the genome is not described by one tree.
- Each genomic position has its own local tree.
- The full collection of local trees + recombination events is the **ARG**.
- ARG inference is much harder than pairwise TMRCA inference.

SINGER

- SINGER adds haplotypes one at a time using **threading**.
- First HMM: sample the **joining branch** along the genome.
- Second HMM: sample the **joining time** conditioned on that branch.
- Then MCMC explores ARG space using **sub-graph pruning and re-grafting (SGPR)**.



SINGER math

- Let B_l be the joining branch at bin *l*.
- SINGER uses branch-specific recombination and rejoining probabilities.
- Transition:

$$P(B_l = b_j \mid B_{l-1} = b_i) = (1 - r_i)\delta_{ij} + r_i \frac{q_j}{\sum_{k:b_k \in S_l} q_k}$$

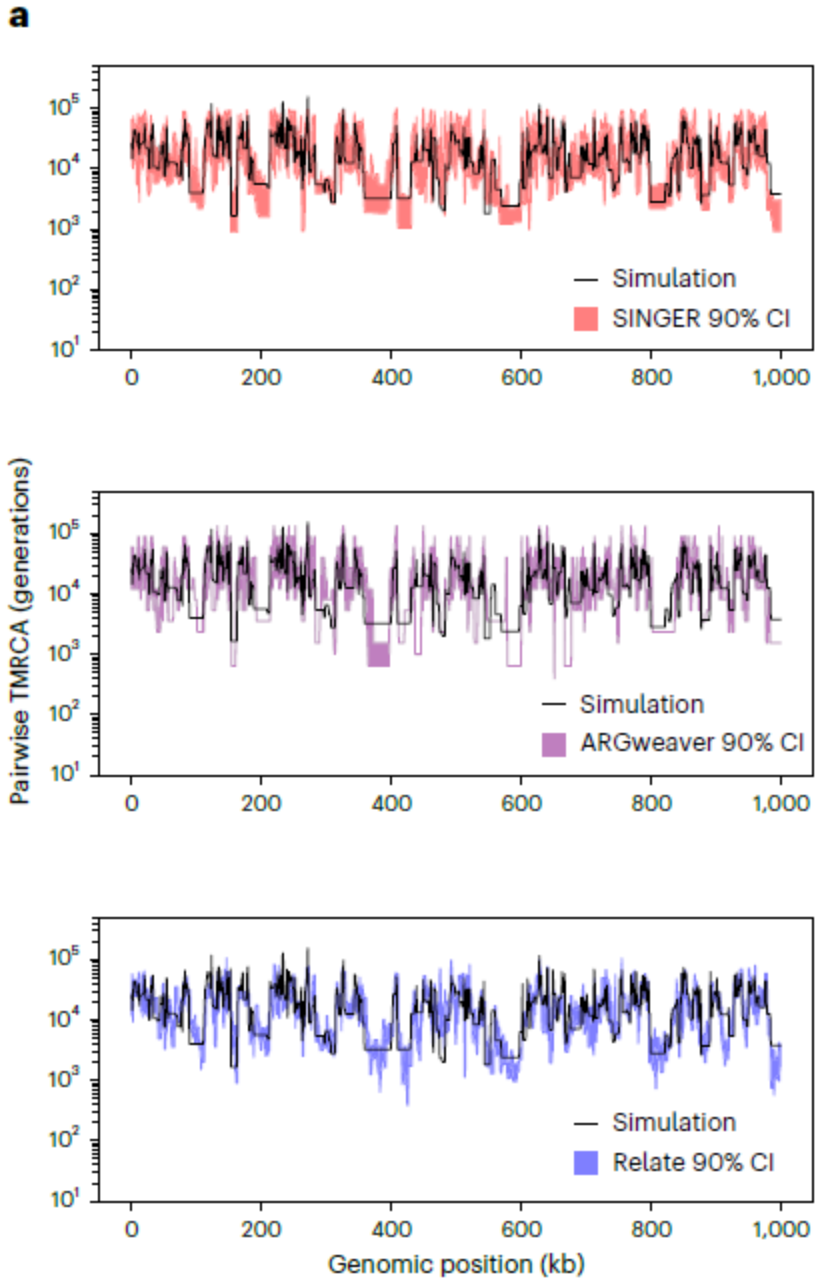
- This keeps computation **linear in the number of hidden states.**

SINGER time sampling and calibration

- After branch sampling, SINGER samples the **joining time** at each bin.
- This step is conceptually similar to **pairwise PSMC**, but now constrained by the chosen branch.
- SINGER also performs **ARG re-scaling** so mutation density matches branch lengths better.
- This improves robustness under model misspecification.

SINGER results

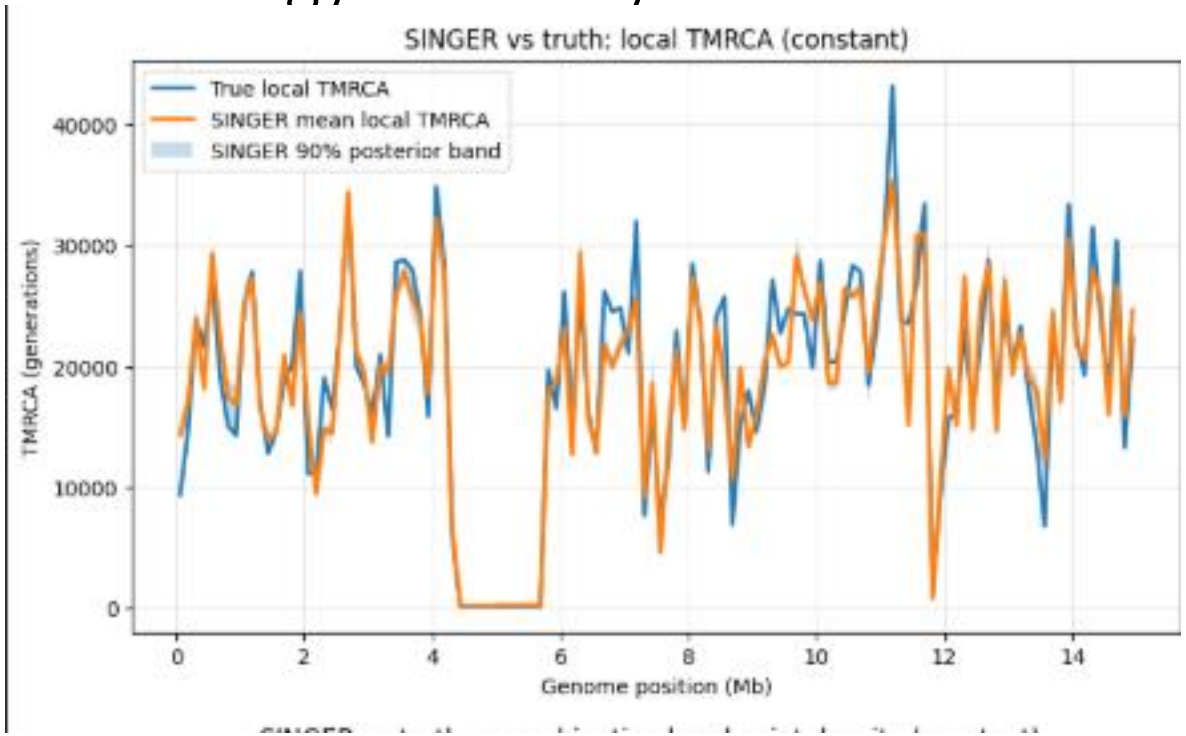
- On simulated data, SINGER had the best **coalescence-time accuracy** among compared methods.
- It also had the best **tree-topology accuracy** on average.
- It is more robust to misspecified N_e and unmodeled size changes.
- Crucially, it gives **posterior samples and uncertainty**, not just one fixed ARG.



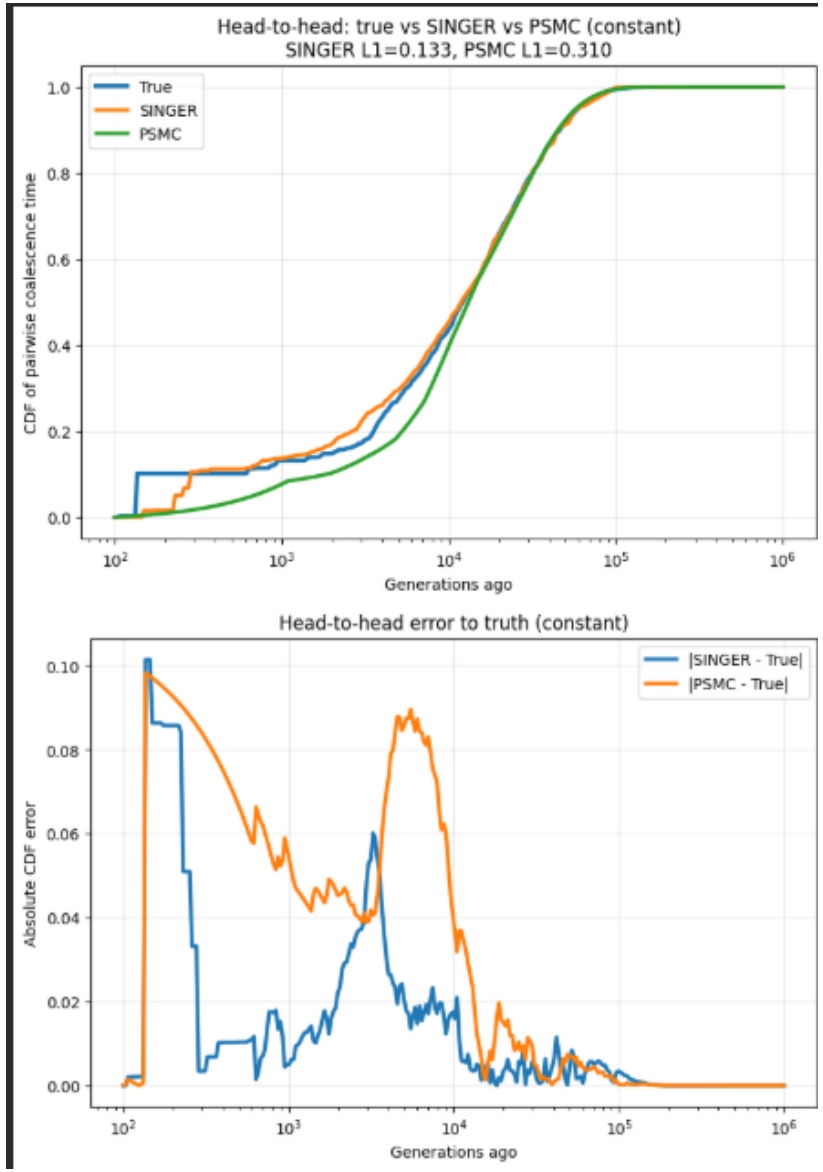
Setup

- We tested **PSMC, SINGER, and PHLASH** on both **msprime-simulated data** and **real human genomes**.
- The simulated data were generated under simple demographic models with recombination, giving us a controlled benchmark with known ground truth.
- The real-data analysis used **one Yoruba, one Chinese, and one European genome** to compare African and non-African demographic histories.
- We then compared the inferred population-size histories, uncertainty patterns, and overall behaviour of the three methods.

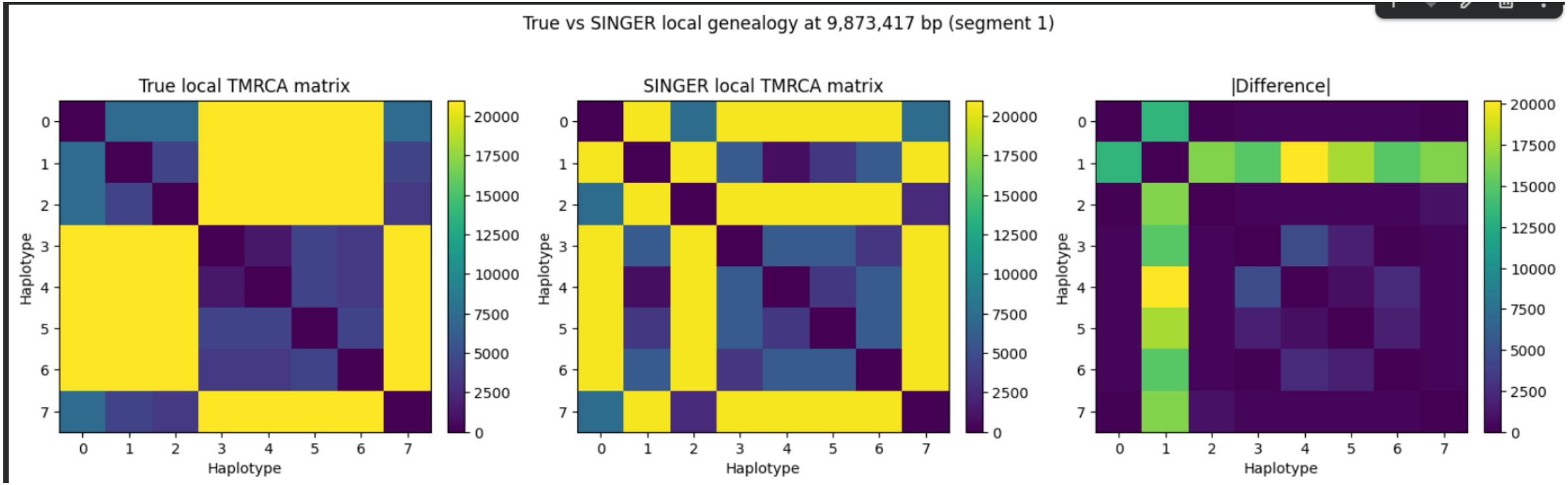
Experiments for SINGER: Comparison with PSMC and Genealogy Recovery



Local genealogy recovery using SINGER.



Global coalescent time distribution: True vs SINGER vs PSMC



Of the 8 Haplotypes, only 1 is mismatched

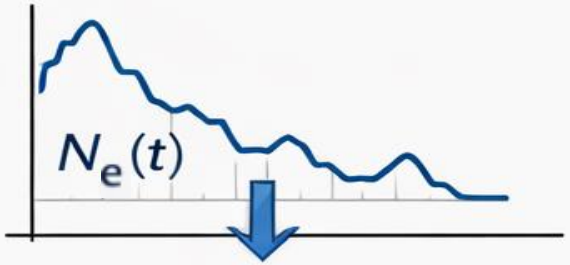
PSMC vs SINGER vs PHLASH:

Conceptual Comparison

Related, but not identical questions.

PSMC

One diploid genome



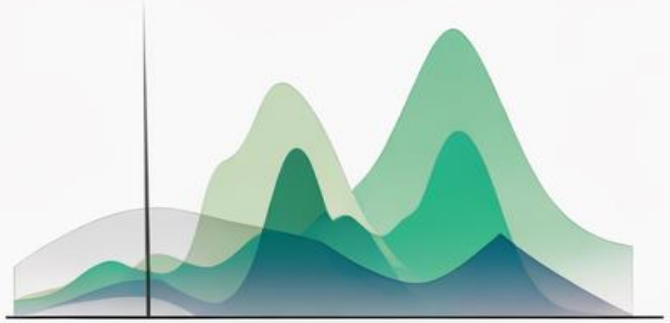
- One diploid genome
- Local pairwise TMRCA
- $N_e(t)$

SINGER

Many phased genomes

- Many phased genomes
- Posterior ARGs
- Topology + Times + Uncertainty

PHLASH



- Many genomes
- Posterior over size histories
- *Scalable Bayesian demography*

Thank you

Any Questions